

Answer all questions in the space provided

Question 1

There are a few methods to synthesize recombinant insulin. One of the methods involves cloning the human insulin cDNA into the cloning vector pUC19 and transforming the clone into *E. coli*.

The modified *E. coli* plasmid vector, pUC19 developed by Fermentas is shown in **Fig. 1.1**. The pUC19 plasmid vector is small (2686bp) with high copy number. The plasmid contains 1) the pMB1 replicon rep (also known as the *Ori* site) responsible for the replication of plasmid, 2) *bla* gene encoding for beta-lactamase that confers resistance to ampicillin and 3) a region of *E. coli* lac operon containing CAP protein binding site, promoter Plac, lac repressor binding site and 5'-terminal part of the lacZ gene encoding the N-terminal fragment of beta-galactosidase. The map shows restriction enzymes that cut pUC19 once.

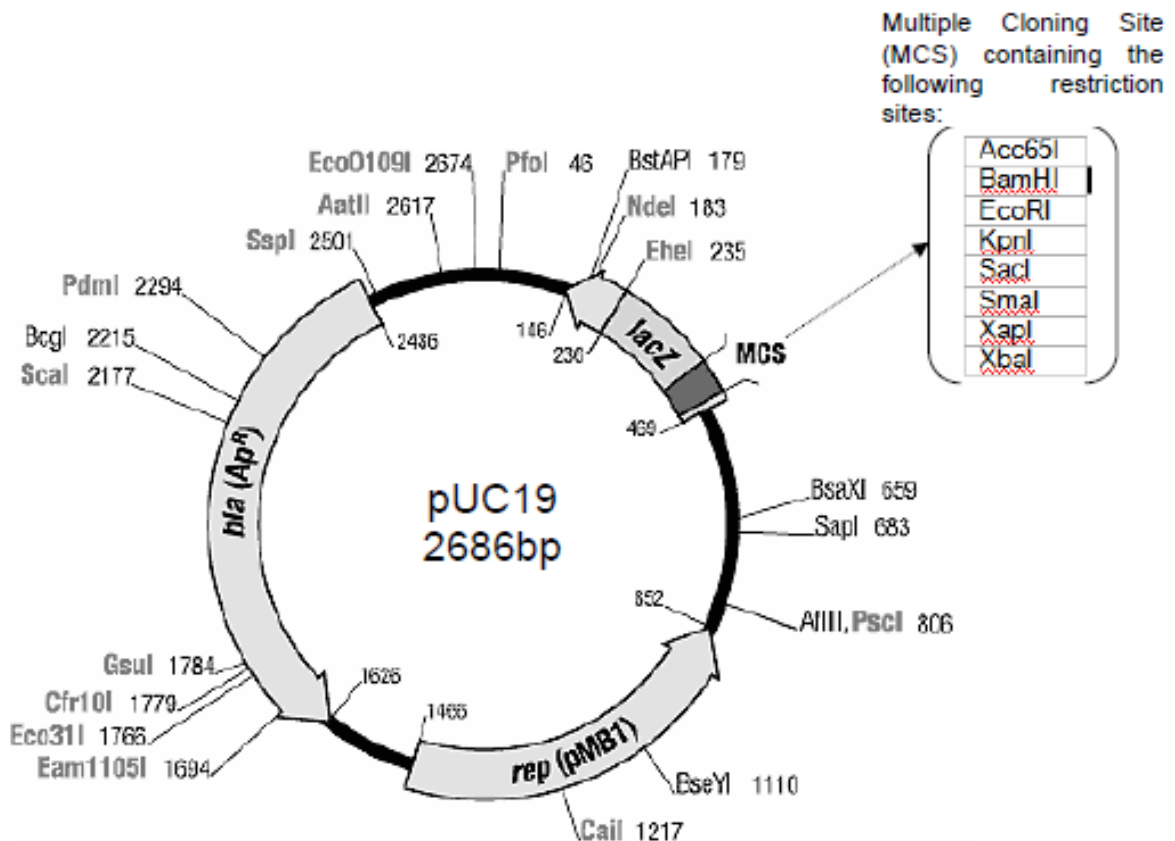


Fig. 1.1

- (a) With reference to **Fig. 1.1**, by means of a labelled arrow, indicate on the diagram the most suitable position to insert the fragment of human insulin cDNA. [1]

anywhere within the lac Z gene (i.e. map position 146 – 469);

- (b) Using the information given in **Table 1.1**, select a restriction enzyme that will generate a blunt end and also allow for subsequent identification of successful recombinant colonies.

(i) State your choice of restriction enzyme: **SmaI** [1]

(ii) Explain your choice of restriction enzyme. [1]

disrupt lacZ gene will allow visual confirmation/ see white colonies (OWTTE);

reject 'selectable marker' (since antibiotic/ bla gene/ amp^R can also be one)

Restriction enzyme	Specific recognition sites	Answer/ elaboration
Acc65I	5'-G [^] G T A C C-3' 3'-C C A T G [^] G-5'	Wrong answer - Within MCS – sticky fragment
BamHI	5'-G [^] G A T C C-3' 3'-C C T A G [^] G-5'	
EcoRI	5'-G [^] A A T T C-3' 3'-C T T A A [^] G-5'	
EheI	5'-T [^] C T A G A-3' 3'-A G A T C [^] T-5'	Wrong answer - Within Lac Z – sticky fragment
KpnI	5'-G G T A C [^] C-3' 3'-C [^] C A T G G-5'	
Pdml	5'-G A A N N [^] N N T T C-3' 3'-C T T N N [^] N N A A G-5'	Wrong answer- Within Amp – blunt fragment (insertion disrupts gene but does not allow for subsequent visual confirmation)
Scal	5'-A G T [^] A C T-3' 3'-T C A [^] T G A-5'	
SmaI	5'-C C C [^] G G G-3' 3'-G G G [^] C C C-5'	Correct answer. 1) blunt fragment 2) within Lac Z gene (allows for subsequent visual identification)

Table 1.1

- (c) Some restriction enzymes generate sticky ends, while others generate blunt ends. Explain why it is more efficient to clone DNA fragments with sticky ends than DNA fragments with blunt ends. [1]

temporary hydrogen bonds to hold 2 fragments (via overhangs) in place
while DNA ligase seals nicks/ phosphodiester bonds are formed/ ligation occurs
(½m each)

- (d) Describe how a set of blunt ended DNA fragment could still be efficiently cloned into a plasmid vector. [3]

- polylinker/ linker;
 - the artificial sequence is double-stranded;
 - cDNA with blunt fragment and vector are cleaved with same restriction enzyme/ or answer in B)i) that generated blunt ends;
 - Ligate polylinker(s) to both cDNA fragment and vector;
 - Linker contains restriction site that can produce fragments with sticky end;
 - Cut/ cleave with same restriction enzymes to generate complementary sticky/overhangs;
 - Add DNA ligase into tube containing both fragments and allow them to ligate and form recombinant DNA molecule;
- (any 3 points)

- (e) Outline the principle of insertional inactivation of the *lacZ* gene. [4]

culture medium contains X-gal, a chromogenic substrate that when cleaved by β -galactosidase, will turn blue;

	<u>Non recombinant</u> cells (½m)	<u>Recombinant</u> cells (½m)
State colour	Blue (½m)	White (½m)
Explain	Complementation of <i>lacZ'</i> gene (in plasmid) and <i>lacZ'</i> gene in <i>E. coli</i> ;	gene of interest disrupts <i>lacZ'</i> gene (in plasmid);

Accept: non-recombinant cells can synthesise functional β -galactosidase/ recombinant cells cannot synthesise functional β -galactosidase (½m)

After transforming the clones into the *E. coli* cells, the cells were plated onto appropriate mediums. Positive results were observed from the morphologies of the grown colonies.

The colonies were harvested and purified via affinity chromatography. A portion of the purified extract was further analysed. Results of the analysis revealed that functional insulin was not present in the extract.

(f) Suggest two reasons why functional insulin was not expressed in the recombinant *E. coli* cells. [4]

1) inserted in wrong direction/ orientation;

resulting in transcription of incorrect strand, hence encoding for a different polypeptide/ or correct/ functional polypeptide is not produced;

2) the α -chain and β -chain of insulin were cloned in separate vectors;

gene of interest was inserted next to lacZ gene therefore fusion polypeptide/ insulin- β -gala polypep was expressed;

functional insulin can only be achieved when the fusion polypeptide is cleaved by CBr/ two separate chains are joined by disulfide bridges;

3) Eukaryotic post-translational mechanisms not present in prokaryotic cells;

No α -chain and β -chain cannot be cleaved and joined to produce functional insulin

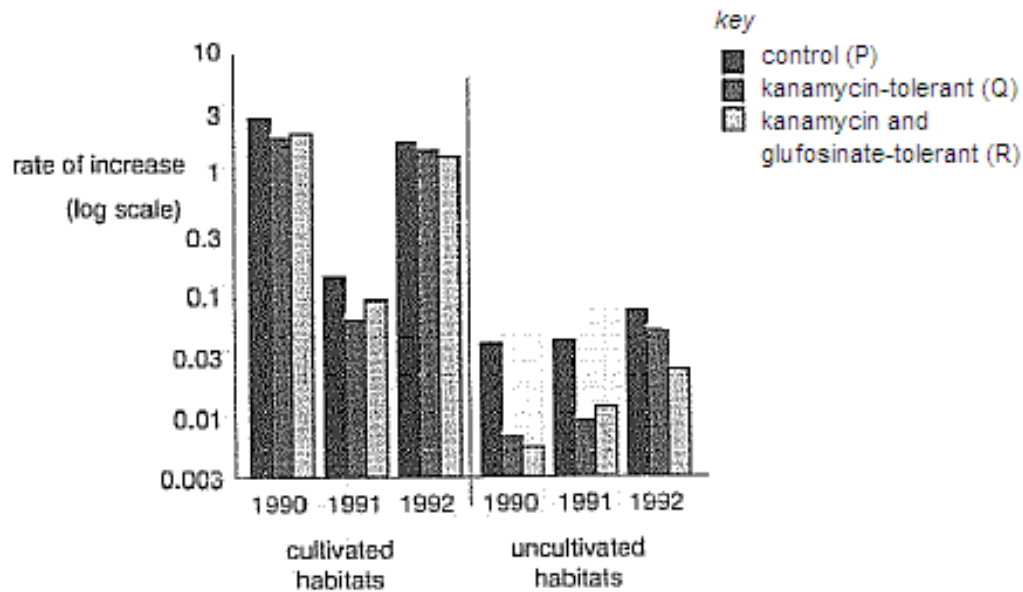
Reject : no transcriptional modification can occur , introns cannot be spliced (already stated in question that cDNA was inserted)

[Total: 15 marks]

Question 2

An experiment was carried out over three years to compare the population growth of normal and genetically engineered oilseed rape plants. Invasiveness was defined as the rate of population increase from one year to the next. Populations in which the rate of increase is greater than 1 would be expected to increase in abundance.

Three different genetic lines (**P**, **Q** and **R**) were compared: non-genetic engineered oilseed rape (**P**) and two genetically engineered versions of the same cultivar. One of these (**Q**) was tolerant of the antibiotic *kanamycin* and the other (**R**) tolerant of both **kanamycin** and the herbicide *glufosinate*. Neither antibiotic nor herbicide was used in the investigation. The rate of population increase of the three genetic lines grown in cultivated and uncultivated habitats are shown in **Fig. 2.1**.



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Fig. 2.1

- (a) Briefly explain how plants are propagated by cloning from tissue culture to obtain the cultivar. [4]
- Small pieces of plant tissues (**explants**) are grown in **culture media**;
 - that typically will contain **sucrose** as a source of energy, **organic nutrients** such as **vitamins and amino acids** and a wide range of **inorganic ion** such as nitrate, **potassium**, phosphate, **magnesium** and trace amounts of copper and iron;;
 - All procedures must be carried out under **aseptic/ sterile conditions**;
 - Plant growth regulators such as **cytokinin and auxin** are added to encourage the cells to divide;
 - The cells divide **mitotically**;
 - to produce an **unorganized/ undifferentiated mass of cells – callus**;
 - callus **differentiate** to form **root in high auxin: cytokinin ratio**;
 - **and shoot in high cytokinin: auxin ratio**;
 - Plantlets are **genetically identical** to one another;
- (b) Tolerance to the antibiotic kanamycin is not expected to give an advantage to the oilseed rape plants. Suggest why this genetic line was included in the investigation. [1]
- **To act as a control so that the actual performance of line R can be accurately assessed**
 - **Or to assess the effect of the procedure of the genetic engineering i.e. to see whether the procedure itself will affect the performance of the plants.**

- (c) With reference to **Fig. 2.1**, compare the invasiveness of non-engineered and genetically engineered oilseed rape plants in cultivated and uncultivated habitats.

(i) Cultivated habitats: [2]

- All 3 lines showed little difference in their rate of increase; QV: for 1990 and 1991, the rate of increase is greater than 1;
- but for 1991, the invasiveness of all 3 lines dropped; below 0.3;
- non-engineered plants was slightly more invasive/ showed the greatest rate of increase throughout the 3 years; QV: for 1991, non-engineered plants is close to 3 while the other 2 are below 3;

(ii) Uncultivated habitats: [2]

- The greatest difference between P and the other 2 lines was shown in 1990 and 1991; QV: P is between 0.03 and 0.1 but both Q and R were below 0.01 on the log scale.
- The differences between them decreased in 1992; with P close to 0.1 and Q between 0.03 and 0.1 and R at 0.03 on the log scale.

OR

- Increase in invasiveness for genetically engineered plant over the 3 years while the control remains more or less constant/ slight increase in invasiveness;; QV showing trend;;
- non-engineered plants were more invasive compared to engineered plants throughout the 3 years; QV;
- all plants have low invasiveness in uncultivated habitats throughout the 3 years/ are less invasive in uncultivated habitats compared to cultivated habitats; QV: all plants below 0.1;

(iii) Suggest an explanation for the observed data. [1]

- With no antibiotic or herbicide being used, there is no selective agent and hence no selective advantage in having the kanamycin or/and glufosinate genes;;
- These probably present themselves as additional genetic material in the cell and hence additional nutrients have to be used to synthesise these gene products leading to a decrease in the growth rate of these plants;;

- (d) Based on the findings of this investigation,

(i) evaluate whether the concern about transgenic crop plants becoming agricultural weeds is justified. [1]

- No major concern/ concern not justified;
- Genetically engineered plant is at most as invasive as the non-engineered plants in both cultivated and uncultivated habitats;;
- QV to support

- (ii) describe 1 potential benefit and 1 hazard of such genetically engineered oilseed rape plants. [2]
- ✓ **Benefit: genetically engineered oilseed rape plants are herbicide-resistant, use of herbicides that control weeds is possible, thus this improves crop yields and productivity.**
 - ✓ **Hazard: Herbicide-resistant genes from GM plants may transfer into wild-life gene pool through cross-pollination, spread resistance to herbicide to weeds, such weeds overrun agricultural areas very rapidly.**
- (e) Explain why plants of the same family as the oilseed rape are at greatest risk of picking up the foreign gene from the genetically engineered plants compared to plants from a different family. [1]
- Oilseed rape from the same family are **genetically similar** and hence genetically compatible with one another;
 - **Pollen** from the genetically engineered plant may be **viable on the stigma** of the other plant/ successful cross-pollination;
 - **pollen tube may germinate, develops pollen tube;**
 - and eventually fuse with the ovum, i.e., **fertilization occurs;**
 - The 2 sets of chromosomes of the hybrid are likely to be **homologous** and are able to **pair up** during meiosis and form **gametes** → fertile hybrids;
 - Ref to sexual reproduction;
 - Ref. to gene flow between individuals of the same family;
 - Ref to no/ less likely to have reproductive barrier between individuals of the same family or specific examples such as flower at the same time of the year, pollinated by same agent, etc.

[Total: 14 marks]

Question 3

A new brightly-coloured mouse was discovered in Russia and it was given the name *Mus colores*. **Fig. 3.1** below shows two flowcharts that illustrate how blue and green cells were isolated from different parts of this mouse, and each cell was cultured individually.

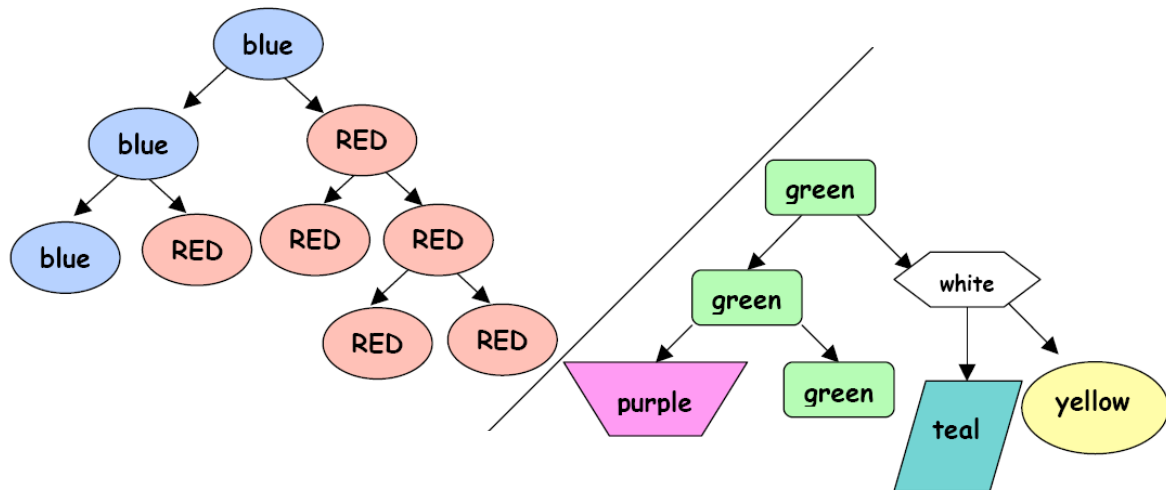


Fig. 3.1

(a) With reference to **Fig. 3.1**,

(i) are the blue cells stem cells? Give a reason for your answer. [1]

Yes as they are capable of self-renew/ giving rise to other cell types.

(ii) given that the green cell is a stem cell, explain why green cells can give rise to differently coloured cells. [2]

Green cell is multipotent. As such, they can give rise to more than one cell type. They can be influenced by the presence of internal and external signals & are capable of asymmetric division which involves unequal distribution of cytoplasmic determinants as well as some genes being switched on/ of. Green cell undergoes differentiation when in contact with neighbouring cells and responds to chemical signals.

Severe combined immunodeficiency disorder (SCID) is an inherited autosomal recessive disorder.

(b) Discuss **one** ethical and **one** social concern regarding the use of gene therapy. [2]

Ethical: Gene therapy may be viewed as tampering with nature and going against the natural way of life;

Social: There may be genetic discrimination in the society as the issue of defining what is a disability/disorder/disease is unclear;

The poor will be discriminated against because they cannot afford the expensive treatment of gene therapy;

(ii) Suggest why a retroviral vector was used. [2]

- retrovirus infect rapidly dividing cells so target stem cells which proliferates;;
- Retrovirus inserts viral DNA into host cell genome hence can incorporate ADA gene into the stem cells;;
- Reference to stability of gene inserted/ longer lasting effect;

(d) Using evidence from Figure 2.1, explain why haematopoietic stem cells in particular are used for gene therapy. [4]

- After gene therapy, total lymphocytes counts and white blood cell counts in peripheral blood increased then maintained throughout the treatment (**trend for A**);
- and **QV** (e.g. Initially, levels were < 200 lymphocytes/ μ l and < 1000 wbc/ μ l respectively before gene therapy but increased to 2000 cells/ μ l and 5000 cells/ μ l respectively by day 150);
- HSCs **differentiate into all blood types** including white blood cells;
- So **lymphocytes**, a type of white blood cells which are **destroyed by toxins which accumulate in ADA deficiency** are also **formed** by the transformed stem cells;
- Increase of transformed T cells and NK cells and levels fairly maintained after gene therapy (**trend for B**);
- And **QV** (e.g. % of transformed cells were 0 at day 0 but increase to 70% and 60% at day 350);
- Showing that the transformed stem cells can give rise to transformed T and NK cells/ reference to HSCs pass the normal allele to daughter cells;
- And levels of transformed cells are increased and maintained at high level after gene therapy → transformed stem cells are a **self-renewing population** of cells;
- May eliminate the need for repeated administrations of the gene therapy/ reference to long-lasting effect of treatment;

[Total: 11 marks]

Planning question

Question 4

One of Africa's most famous extinct animals, the quagga, was a subspecies of the plains zebra, which was once found in great numbers in South Africa's Cape Province and the southern part of the Orange Free State.

It was distinguished from other zebras by having the usual vivid marks on the front part of the body only. In the mid-section, the stripes faded and the dark, inter-stripe spaces became wider, and the hindquarters were a plain brown.

Hunters in Africa have caught an animal which resembles a quagga. However, they could not be sure.

Plan an experiment to investigate if this animal which the hunters caught is a descendant of the quagga.

Your planning must be based on the assumption that you have been provided with the following equipment and materials

- Tissue samples from museum specimens of a real quagga and from the animal that the hunters caught
- Pestle and mortar
- DNA extraction buffer solution
- Glass rods
- Microcentrifuge tubes
- Centrifuge
- Restriction enzyme
- Agarose or polyacrylamide gel plate
- Suitable source of electrical current
- Radioactive probe
- Nitrocellulose membrane
- Autoradiography equipment

Your plan should have a clear and helpful structure to include

- An explanation of the theory to support your practical procedure
- A description of the method used, including the scientific reasoning behind the method
- The type of data generated by the experiment
- How the results will be analyzed including how the origin of the organism can be determined

Mark Scheme

1. Theoretical consideration or rationale of the plan to justify the practical procedure;
2. Method of DNA extraction including homogenization and use of buffers;
3. Preparation of samples for electrophoresis, including use of centrifuge;
4. Selection of restriction enzyme and reasons for the selection;
5. Amplification of DNA fragments using PCR including detail of PCR;
6. Separation of fragments by gel electrophoresis and the principles behind the separation;
7. Transfer of DNA onto nitrocellulose membrane;
8. Hybridization with radioactive labeled DNA probe;
9. Autoradiography method;
10. Method of band visualization, e.g. exposure to X-ray or chemiluminescent substrate;
11. Significance of matching bands;
12. The correct use of technical and scientific terms;
13. Bonus: valid safety concerns

Free-response question

Write your answer on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Question 5

- (a) Describe the different types of stem cells and using named examples, explain their normal functions in a living organism. [6]

1. totipotent embryonic stem cells, unspecialized;
2. from fertilized egg into 8-cell stage embryo develops into blastocyst via mitosis;
3. retains complete set of genetic information;
4. can give rise to any cell type / entire organism;
5. pluripotent embryonic stem cells from inner cell mass of blastocysts;
6. capable of continually renewing and dividing through cell division for long periods of time;
7. develops into three primary germ layers, ultimately giving rise to entire organism;
8. multipotent (adult) stem cells important in developing tissues that give rise to multiple specialised cell types;
9. multipotent haemopoietic stem cells of adult stem cells;
10. giving rise to all types of blood cells e.g. erythrocytes, B lymphocytes, T lymphocytes, neutrophils, basophils, platelets;
11. unipotent (adult) stem cells give rise to new cells of the same specific type, e.g. liver cells
12. small number of adult stem cells found in various parts of the body, e.g. in brain / bone marrow / blood vessels / skeletal muscle / skin / liver;
13. adult stem cells important as they generate replacements for cells that are lost in disease, tissue injury & normal wear-and-tear;
14. neuronal stem cells develops into neurons / nerve cells / skin stem cells develops into skin nerve cells / bone marrow stem cells develops into blood cells, liver cells, muscle cells;

- (b) Describe how cystic fibrosis can be cured using gene therapy and the technical challenges scientists still face when designing such a cure. [8]

[Note: there are two parts to this question – 1) How it is cured and the challenges faced.]

Part 1 - Cure using genetic therapy (students have to be concise and state that the corrective allele is introduced; what defect this overcomes and how the gene is introduced)

1. Genetic disease due to a deletion mutation of 3 nucleotides on chromosome 7, resulting in the loss of phenylalanine in polypeptide. It follows an autosomal recessive inheritance pattern; [1] *A mark was given for the genetic basis of the disease as this sets the basis for the gene therapy*
2. Scientist use either viral (Adenovirus) or non-viral delivery (liposomes) methods to introduce a functioning CFTR allele into the human;
3. The normal, functional allele coding for CFTR found on chromosome 7 isolated from a normal individual;

4. E.g., Lipofection - A aerosol spray or dispenser/nebulizer contain the liposome with the corrective CFTR allele can be used to target the lung epithelial cells;
5. E.g., Alternative viral delivery system for Cystic Fibrosis – Adenovirus that can target the cells;
6. Successful therapy results in the CFTR protein being expressed and integrated into the cell membrane
(max 4)

Part 2 - Challenges

7. Treatment is transient as epithelial cells are constantly being shed, so repeated therapy is needed;
8. Cystic fibrosis affects other organs that are less accessible. currently cure is targeted at the lungs, but the pancreas is also affected so scientists have to come up of a way to target the epithelial cells of the pancreas
9. Also, difficult to carry out *ex vivo* and test for successful incorporation of the corrective allele since the problem is with epithelial cells
10. Difficult to target the basal cells of the epithelium of every organ;
11. Viral delivery methods are specific and ensure effectiveness since the gene may be incorporated into the genome or inserted into the nucleus but may be harmful as the immune response may be stimulated
12. Liposomes have low transfection efficiency as they do not have specificity and infection mechanisms that viruses have
13. Liposomes have low integration efficiency – have to specifically engineer to include regions of homology, DNA will not integrate into host cell's genome
14. Liposomes have a small degree of toxicity – so many rounds of treatment can be harmful to the cells or tissues being treated
(max 4)

- (c) With reference to a named example, explain how RFLP can be used in disease detection. [6]

- **NAMED DISEASE**
- sickle-cell anaemia

- **BACKGROUND ON α -GLOBIN GENE**
- mutant allele
- is caused by point mutation / base substitution
- within a Ddel restriction site
- in the 5' end of the human β -globin gene

- **RFLP ANALYSIS**
- Ddel recognises and cuts the specific DNA sequence 5' CTNAG 3'
- in the mutant allele, the point mutation prevents the restriction enzyme from recognising and cutting that particular restriction site
- cutting a human DNA sample with the Ddel restriction enzyme will produce a different mixture of restriction fragments

- for the normal and mutant allele respectively
- each allele will have its unique band pattern in gel electrophoresis
- reference to 'single nucleotide polymorphisms'
- that help to distinguish the normal and sickle-cell alleles of the α -globin gene
- reference to use of a radioactively-labelled, single-stranded DNA probe
- complementary to β -globin gene
- to detect / identify the restriction fragments bearing α -globin gene sequences
- restriction fragment arising from the normal α -globin allele is shorter than that arising from the mutant β -globin allele
- reference to homozygotes having single bands for the normal and mutant alleles respectively
- reference to heterozygotes having more than 1 band, i.e. arising from both normal and mutant alleles